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REGRESSION AND NEUTRAL NETWORKS

Random state	Regression alg1	Regression alg2	NN
0	1.0	1.0	0.34
1	1.0	1.0	0.35
2	1.0	1.0	0.31
3	1.0	1.0	0.23
4	1.0	1.0	0.31
5	1.0	1.0	0.34
6	1.0	1.0	0.21

Mean	**1.00**	**1.00**	**0.299**
Lowest	**1.00**	**1.00**	**0.21**
Highest	**1.00**	**1.00**	**0.35**

INTRODUCTION

In this task, THERE ARE three different models , were used to predict housing prices: two traditional regression models and a neural network. The performance of each model was evaluated using the **R^2 score**, which indicates how well the model fits the data. A perfect R^2 score of 1.0 means the model perfectly predicts the target variable.

From the table above, we observe that both regression models (Regression Algorithm 1 and 2) consistently achieved an R^2 score of **1.0** for every random state. This suggests that the models performed extremely well on the given dataset t. In contrast, the **neural network NN** showed much lower and more variable performance, with R^2 scores ranging from **0.21 to 0.35**, indicating that the NN struggled to capture the underlying patterns in the data.

Explanation of Model Functionality

Regression Algorithms (Alg1 and Alg2):

Linear Regression (Regression Alg 1):
Linear regression is a basic statistical method used for predicting a dependent variable (in

this case, housing prices) based on one or more independent variables. The algorithm assumes a linear relationship between the target and input variables, finding the best-fitting line that minimizes the sum of squared residuals. **Reference**: Introduction to Statistical Learning by James (2013).

Decision Tree Regression (Regression Alg 2):
Decision trees are non-linear models that split the data into subsets based on the input features, aiming to minimize the error within each subset The tree is built by selecting the feature that best splits the data, recursively splitting until a stopping criterion is met

Reference: Pattern Recognition and Machine Learning by Bishop (2006).

Neural Network (NN):

A neural network is a machine learning model inspired by the human brain's architecture. It consists of layers of neurons where each neuron receives inputs, processes them with an activation function and sends the output to the next layer. Neural networks are powerful for capturing complex, non-linear relationships, making them suitable for problems with large, complex datasets. **Reference**: Deep Learning by Goodfellow, (2016).

Suggestions for Model Improvements .

Improving the Regression Models (Alg 1 & 2)

Both regression algorithms achieved an R^2 score of 1.0, suggesting overfitting, where the models memorize the training data and fail to generalize to unseen data .
to solve:

We Introduce **regularization** to reduce overfitting. For Linear Regression, we could use **Ridge Regression**, which penalize large coefficients. For Decision Trees, we can adjust the **maximum depth** or use **Random Forest** to average the results of multiple trees, which reduces overfitting.

Improving the Neural Network

The neural network only trained for 7 epochs, which is likely insufficient for it to capture the patterns in the data. Training for more epochs might improve its performance.

To solve this:
Increase the number of epochs to 50 or 100 and monitor the model's performance over time.

Conclusion

In conclusion, while the traditional regression models (Linear and Decision Tree Regression) performed very well, showing perfect results ($R^2 = 1.0$), they likely suffered from overfitting. The neural network, on the other hand, struggled with lower R^2 scores, but this can be improved by increasing the epochs, adjusting the model architecture, and fine-tuning hyperparameters.